



Fig. 7: Testing with keyboard and voice output from speaker

Simply ask a question, seek an answer, or request a code snippet, and the ESP32 will assist in writing the problem statement on the TFT screen using a PS/2 keyboard connected to the ESP32. The ESP32 will then send the question to ChatGPT, and the output obtained from ChatGPT will be displayed on the same TFT. Simultaneously, the loudspeaker connected to the I2S amplifier (MAX98357A) will speak out the text visible on the screen.

No multimedia computer is needed for this work, though an

internet Wi-Fi connection and a secret key to access the ChatGPT API are necessary. This secret key is the single point of access for the openai.com API; no other login/password is required.

You might wonder why an old PS2 keyboard is used instead of a USB keyboard. Although an attempt was made to use a sleek, small USB keyboard, it could not communicate with the ESP32. The same applies to the sleekest Bluetooth keyboard.

The software code is developed in the Arduino IDE. In the code, it is necessary to configure the OPEN AI API and the Wi-Fi SSID and password. After configuring the code, the source code can be uploaded to the ESP32 by selecting the correct COM port and board as ESP32. Fig. 6 displays a snippet of the source code.

Construction and testing

You may assemble the circuit on a general-purpose PCB, as shown in the author's prototype. First, upload the source code of the ter-

minal into the ESP32 board. Then, refer to Tables 2 through 4 before assembling. After powering on, wait for some time.

The delays used inside loops are quite specific. You may adjust them, but it is suggested to start with the provided values. Once an understanding of the responses is gained, adjustments can be made, as needed. The initial questions, such as 'who are you?', were answered meticulously by the ChatGPT, producing a self-introduction on the screen, and the speaker delivering it nicely.

Subsequently, more advanced questions were asked, such as 'talk about EFY' and 'the distance between Earth and the Sun.' Each time, the ChatGPT understood the narrative clearly and provided meticulous answers. The speaker worked flawlessly to deliver the voice output clearly and loudly.

The most advanced level of questioning involved tasks like writing 5 sentences about EFY, India, NTPC, Google.com, etc. Throughout all the tests, the ChatGPT performed exceptionally well.

However, as these answers exceeded 200 characters, Google TTS refused to speak them. To address this, the answer string was modified to trim it to 200 characters. This way, while the full answer appears on the screen, internally it speaks up to 200 characters only. The author's final prototype, including the keyboard used in testing, is shown in Fig. 7.

Aftermath. The next modification for this terminal will make it fully voice interactive, so it will listen to questions and respond like an obedient robot! This development is for the EFY readers. **EFY**

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com



Somnath Bera is an electronics enthusiast. He has contributed many articles across the globe as a freelancer.